

## GOIANIA, Brazil

WMO: 834240

Lat: 16.632S

Lon: 49.221W

Elev: 747

StdP: 92.67

Time Zone: -3.00 (SAB)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |     |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-----|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99% |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP  | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g) | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 6             | 12.9       | 14.0 | 1.2                           | 4.5 | 30.9 | 3.2 | 5.2 | 29.3 | 8.1                   | 25.4 | 7.3 | 25.4 | 1.2                   | 130  | 0.293 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 9             | 13.9                   | 35.9            | 19.1 | 34.8 | 19.3 | 33.9 | 19.8 | 24.5                | 29.8 | 24.1 | 29.3 | 23.7 | 28.8 | 3.4                  | 330  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 23.1                            | 19.6 | 26.1 | 22.9 | 19.3 | 25.9 | 22.2 | 18.5 | 25.3 | 78.3          | 29.8 | 76.8 | 29.5 | 75.4 | 28.9 | 28.8              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|
| 1%                | 2.5% | 5%  | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |
|                   |      |     | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
| 7.8               | 6.8  | 5.8 | 9.5                        | 37.4 | 1.7                | 1.4 | 8.3                                                | 38.4 | 7.3        | 39.2 | 6.4        | 40.0 | 5.2        | 41.0 |
|                   |      |     | 8.0                        | 25.1 | 1.5                | 1.6 | 6.9                                                | 26.3 | 6.0        | 27.2 | 5.2        | 28.1 | 4.1        | 29.3 |
|                   |      |     | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |
|                   |      |     | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 24.9   | 25.6  | 25.9  | 25.7  | 25.2  | 23.5  | 22.4  | 22.4  | 24.2  | 26.6  | 26.7  | 25.8  | 25.5 |
|                                                                                  | DBStd       | 2.09   | 1.38  | 1.31  | 1.26  | 1.34  | 1.48  | 1.54  | 1.71  | 1.67  | 1.71  | 1.82  | 1.57  | 1.32 |
|                                                                                  | HDD10.0     | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |
|                                                                                  | HDD18.3     | 3      | 0     | 0     | 0     | 0     | 1     | 1     | 1     | 0     | 0     | 0     | 0     | 0    |
|                                                                                  | CDD10.0     | 5457   | 483   | 444   | 488   | 457   | 418   | 371   | 384   | 439   | 498   | 518   | 473   | 482  |
|                                                                                  | CDD18.3     | 2418   | 225   | 211   | 230   | 207   | 161   | 122   | 127   | 181   | 248   | 259   | 223   | 224  |
|                                                                                  | CDH23.3     | 21470  | 1642  | 1602  | 1681  | 1663  | 1380  | 1196  | 1427  | 2143  | 2908  | 2591  | 1702  | 1536 |
| CDH26.7                                                                          | 9086        | 629    | 646   | 642   | 658   | 489   | 402   | 546   | 1016  | 1566  | 1276  | 672   | 546   |      |
| Wind                                                                             | WSAvg       | 2.3    | 2.3   | 2.1   | 2.0   | 2.0   | 2.0   | 2.1   | 2.4   | 2.8   | 2.9   | 2.6   | 2.4   | 2.4  |
| Precipitation                                                                    | PrecAvg     | 1571   | 253   | 236   | 254   | 107   | 29    | 8     | 2     | 11    | 49    | 130   | 220   | 272  |
|                                                                                  | PrecMax     | 1908   | 486   | 391   | 526   | 243   | 75    | 70    | 15    | 52    | 121   | 294   | 313   | 474  |
|                                                                                  | PrecMin     | 1088   | 81    | 129   | 94    | 0     | 0     | 0     | 0     | 0     | 2     | 22    | 117   | 158  |
|                                                                                  | PrecStd     | 202    | 98    | 63    | 90    | 68    | 22    | 15    | 3     | 14    | 32    | 70    | 56    | 88   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 34.1  | 34.2  | 33.8  | 33.8  | 32.2  | 31.9  | 32.8  | 35.0  | 37.0  | 37.2  | 35.2  | 33.7 |
|                                                                                  |             | MCWB   | 21.0  | 22.6  | 22.7  | 22.8  | 20.6  | 18.6  | 18.4  | 18.6  | 19.0  | 18.7  | 21.3  | 22.2 |
|                                                                                  | 2%          | DB     | 32.9  | 33.1  | 32.5  | 32.2  | 31.2  | 30.9  | 31.6  | 33.9  | 36.0  | 35.9  | 33.2  | 32.2 |
|                                                                                  |             | MCWB   | 21.3  | 22.4  | 22.5  | 21.8  | 20.5  | 19.0  | 17.7  | 18.0  | 18.7  | 19.0  | 21.5  | 21.8 |
|                                                                                  | 5%          | DB     | 31.8  | 32.0  | 31.5  | 31.2  | 30.2  | 30.0  | 30.9  | 32.9  | 34.9  | 34.3  | 31.9  | 31.1 |
|                                                                                  |             | MCWB   | 21.7  | 22.1  | 22.4  | 21.7  | 20.2  | 18.7  | 17.7  | 17.6  | 18.7  | 19.8  | 21.7  | 22.2 |
|                                                                                  | 10%         | DB     | 30.2  | 30.8  | 30.2  | 30.2  | 29.6  | 29.1  | 29.9  | 31.9  | 33.9  | 33.0  | 30.7  | 29.9 |
|                                                                                  |             | MCWB   | 22.1  | 22.2  | 22.4  | 21.7  | 20.0  | 18.5  | 17.5  | 17.3  | 18.6  | 20.2  | 21.7  | 22.3 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 24.6  | 25.2  | 24.8  | 24.4  | 23.5  | 21.7  | 20.8  | 21.4  | 23.3  | 24.0  | 24.4  | 24.6 |
|                                                                                  |             | MCDB   | 29.7  | 30.6  | 30.1  | 30.2  | 29.0  | 28.6  | 29.2  | 30.1  | 31.1  | 30.5  | 29.9  | 29.5 |
|                                                                                  | 2%          | WB     | 24.0  | 24.4  | 24.2  | 23.8  | 22.6  | 20.7  | 19.8  | 20.0  | 22.5  | 23.4  | 23.7  | 24.0 |
|                                                                                  |             | MCDB   | 29.0  | 29.5  | 29.4  | 29.4  | 28.1  | 27.4  | 28.4  | 28.5  | 29.9  | 29.6  | 28.9  | 28.8 |
|                                                                                  | 5%          | WB     | 23.5  | 23.9  | 23.8  | 23.3  | 21.8  | 20.1  | 19.0  | 19.1  | 21.8  | 22.8  | 23.2  | 23.5 |
|                                                                                  |             | MCDB   | 28.2  | 28.8  | 28.7  | 28.7  | 27.7  | 26.9  | 27.9  | 29.0  | 28.8  | 28.8  | 28.2  | 28.0 |
|                                                                                  | 10%         | WB     | 23.1  | 23.4  | 23.4  | 22.9  | 21.1  | 19.4  | 18.4  | 18.5  | 21.0  | 22.3  | 22.8  | 23.1 |
|                                                                                  |             | MCDB   | 27.5  | 28.0  | 28.0  | 27.9  | 26.5  | 26.3  | 27.2  | 28.8  | 27.4  | 28.0  | 27.6  | 27.2 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 9.1   | 9.7   | 9.4   | 10.6  | 12.2  | 13.7  | 14.8  | 15.0  | 13.9  | 11.4  | 9.5   | 8.8  |
|                                                                                  |             | MCDBR  | 11.9  | 11.8  | 11.3  | 11.9  | 13.4  | 15.1  | 16.3  | 16.8  | 16.3  | 14.3  | 11.9  | 11.0 |
|                                                                                  |             | MCWBR  | 3.5   | 3.6   | 3.4   | 3.7   | 4.6   | 5.1   | 5.2   | 4.8   | 4.3   | 3.4   | 3.3   | 3.3  |
|                                                                                  | 5% WB       | MCDBR  | 9.0   | 9.8   | 9.5   | 10.3  | 11.6  | 13.4  | 14.6  | 15.1  | 12.2  | 10.6  | 9.4   | 8.7  |
|                                                                                  |             | MCWBR  | 3.4   | 3.6   | 3.5   | 3.7   | 4.5   | 5.1   | 5.5   | 5.0   | 3.8   | 3.5   | 3.4   | 3.2  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.391  | 0.389 | 0.380 | 0.365 | 0.341 | 0.328 | 0.339 | 0.377 | 0.491 | 0.468 | 0.414 | 0.404 |      |
|                                                                                  | taud        | 2.450  | 2.447 | 2.489 | 2.502 | 2.539 | 2.574 | 2.500 | 2.360 | 2.056 | 2.186 | 2.364 | 2.404 |      |
|                                                                                  | Ebn at Noon | 953    | 946   | 932   | 911   | 900   | 896   | 893   | 885   | 818   | 864   | 925   | 941   |      |
|                                                                                  | Edh at Noon | 122    | 121   | 112   | 104   | 93    | 87    | 96    | 118   | 170   | 155   | 132   | 127   |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 5.63   | 5.76  | 5.49  | 5.43  | 4.97  | 4.75  | 5.03  | 5.69  | 5.84  | 5.83  | 5.40  | 5.49  |      |
|                                                                                  | RadStd      | 0.63   | 0.39  | 0.41  | 0.33  | 0.20  | 0.18  | 0.21  | 0.30  | 0.47  | 0.49  | 0.37  | 0.46  |      |

## Historical Trends

|                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| Station Only           | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| Regional (2 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
|                        | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page